

Roll No.....

End Semester Examination

Name of the Program: B.Tech (ECE)

Semester: VI

Name of the Course: Wireless Communication

Course Code: TEC 601

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Draw the seven-cell reuse pattern and verify that the cluster size is $N = i^2 + j^2 + ij$.	
(b)	(i) The service area in a cellular system requires 12 repetitions of a cluster having 7 cells. Each cell has 16 channels. Find the total number of channels that can be simultaneously accessed in the system. (ii) Discuss different handoff schemes based on the direction.	CO1
(c)	A cellular system experiences path loss exponent to be 4, where a UE requires minimum 18 dB SIR to have acceptable voice quality. If there 7 cells in a cluster, will it support the minimum required SIR? Now, assume the minimum required SIR is increased to 20 dB. Find the appropriate cluster size.	
Q2	(20 marks)	
(a)	(i) A cellular system has a total of 32 cells, with radius of 1.6 km for each cell, using 7 cells in a cluster. It has been assigned a total of 336 duplex channels to handle the voice traffic. What is the area covered by the cellular system? Find the number of channels per cell and state how many simultaneous calls are possible over the cellular system. (ii) Explain meaning of the terms 'universal frequency reuse' and 'fractional frequency reuse'.	CO1 & 2
(b)	State significance of two-ray ground reflection model. Explain using suitable diagrams and obtain an expression for the received signal power.	
(c)	Assuming 7 cell cluster, find an expression to indicate SIR in co-channel cells in the downlink.	
Q3	(20 marks)	
(a)	What is the difference between rms delay spread and maximum excess delay for the multipath power delay profile? Explain.	CO 3
(b)	What do you mean by Doppler shift and Doppler spread? Use suitable mathematical derivations to explain.	
(c)	Assume a mobile equipment operates over 900 MHz frequency and moves with a velocity of 70 kmph. Find the received signal frequency if the mobile is moving (a) directly towards the transmitter, (b) directly away from the transmitter, and (c) in a direction which is perpendicular to the direction of arrival of the transmitted signal	
Q4	(20 marks)	
(a)	How does Selection Combining function? Derive an expression to represent mean SNR improvement realized in Selection Combining.	CO4
(b)	What do you mean by Rayleigh & Rician fading Model? Derive an expression to represent PDF of fading amplitude under Rayleigh fading Model.	
(c)	(i) Explain under what conditions slow fading and fast fading occur. (ii) Explain different types of fading.	

Q5	(20 marks)	CO5 & CO6
(a)	Discuss the processes involved in direct sequence spread spectrum (DSSS) and frequency hop spread spectrum (FHSS).	
(b)	State and explain the properties of maximum length sequence?	
(c)	What is spread spectrum technique? How spreading and de-spreading are performed?	